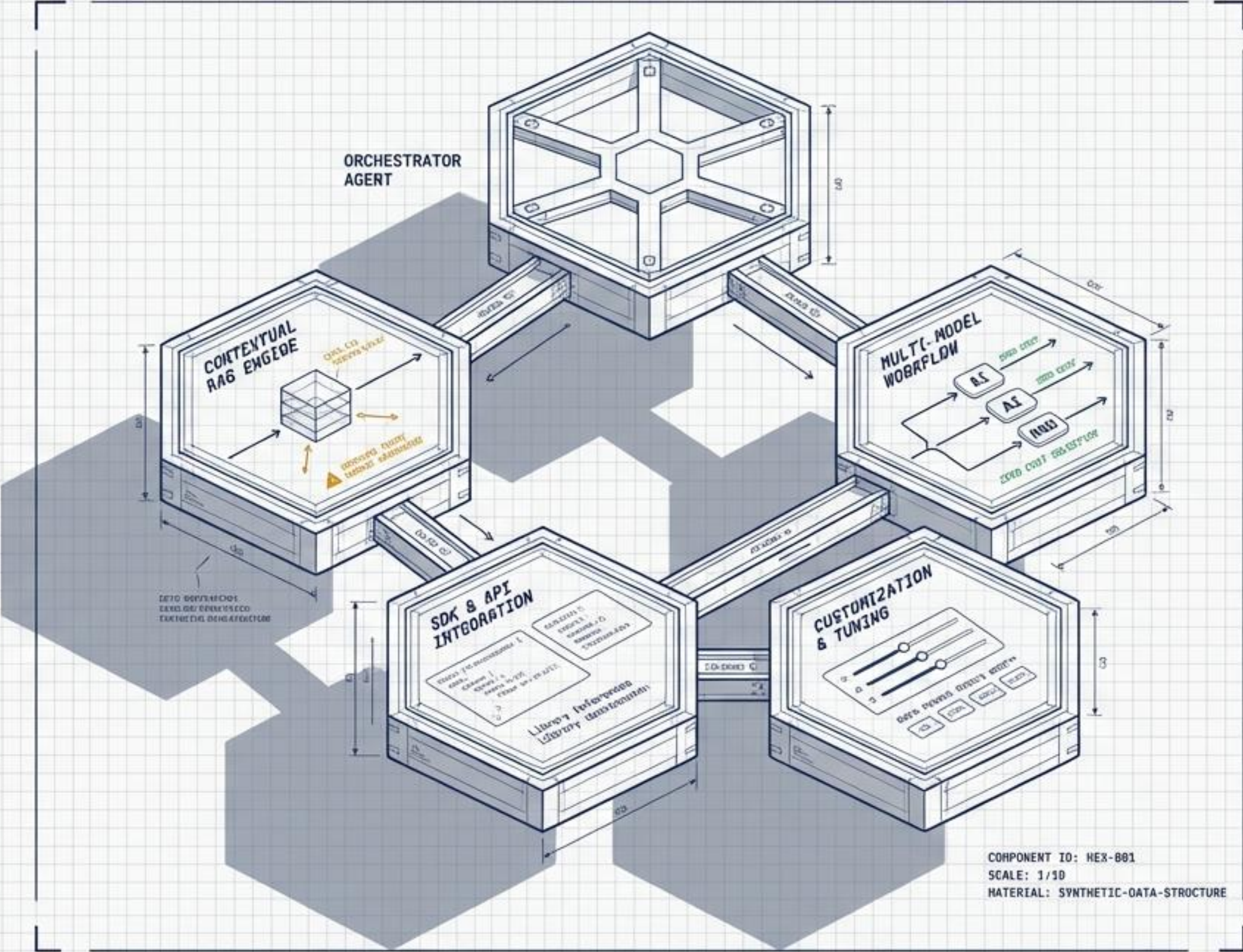
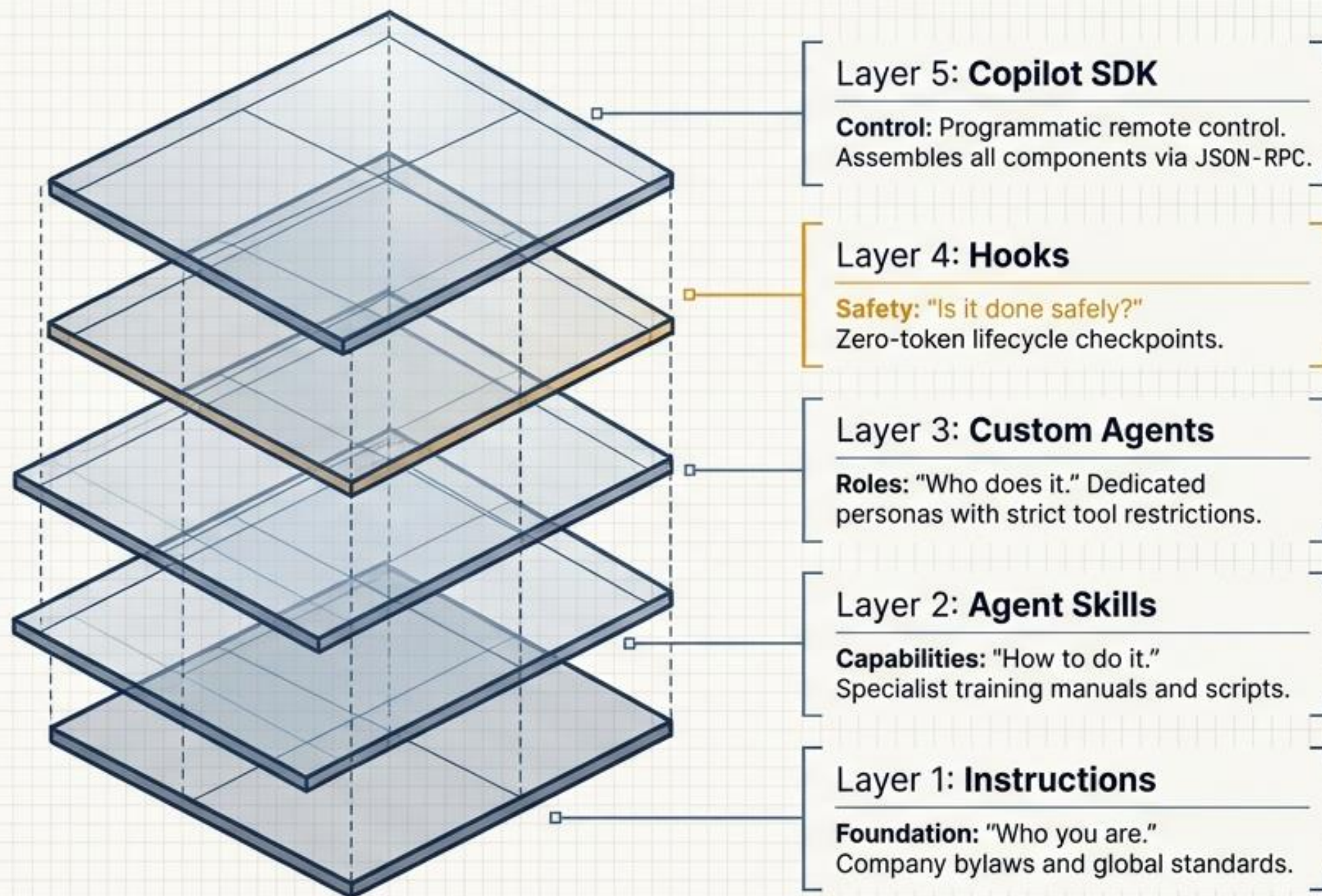


GitHub Copilot Customization Architecture

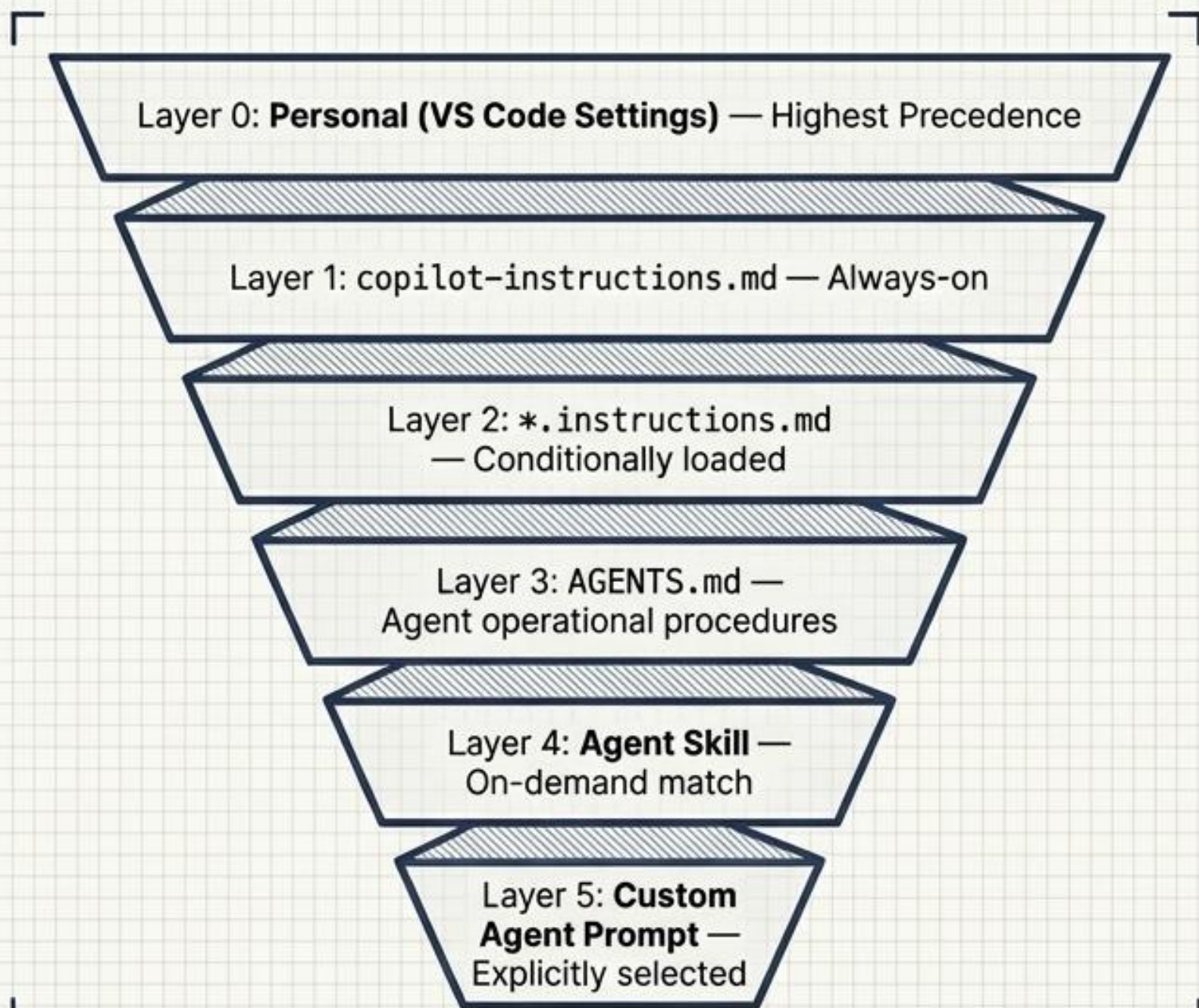
The Principal Engineer's Guide to Multi-Agent Workflows, Token Strategy, and SDK Integration.



THE FIVE-PILLAR ARCHITECTURE



THE CONTEXT FUNNEL AND PRECEDENCE HIERARCHY



>_	>_
<code>copilot-instructions.md</code>	<code>applyTo: '*'</code>
Unconditional trigger. Injected into every chat , agent run , and review . Rule: Keep to 5-15 core rules max.	Conditional trigger. Requires active file context . Can excludeAgent (e.g., exclude from code-review).

AGENT SKILLS AND PROGRESSIVE DISCLOSURE

TIER 1: METADATA (ALWAYS-ON)

The name and description YAML frontmatter. Costs ~50-100 tokens.

TIER 2: BODY (ON MATCH)

The SKILL.md body content. Loads only when the task matches the description. Limit: < 500 tokens.

TIER 3: REFERENCES (ON DEMAND)

Files stored under /references/. Loads only when explicitly referenced. Limit: < 1,000 tokens per file.

ARCHITECTURAL INSIGHT:

The description field is the sole criterion Copilot uses to load a skill.
Precision here is your highest-leverage optimization.

DEFINING TEAMMATES WITH CUSTOM AGENTS

Planner Agent

Tools Whitelist:
['search', 'fetch']
(Read-only)

Handoff Target:
Implementer

Implementer Agent

Tools Whitelist:
['editFiles',
'terminalLastCommand']
(Write-capable)

Sub-agents:
['Researcher']

Security Reviewer

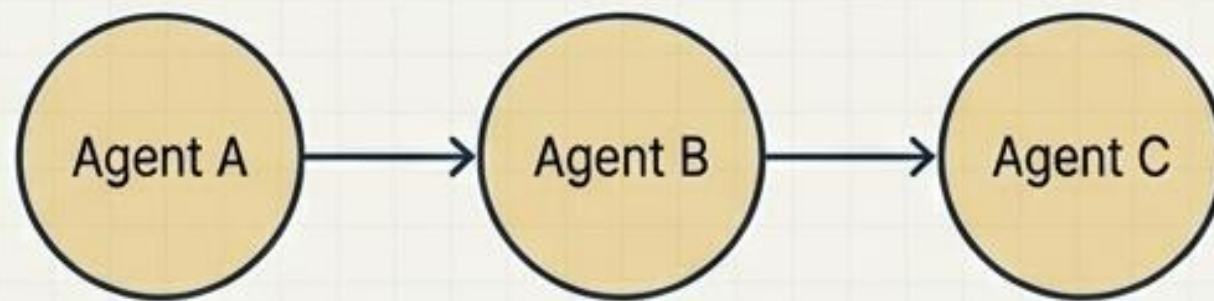
Tools Whitelist:
['codebase']
(Read-only)

Model Isolation:
Claude Opus 4.5

Principle of Least Privilege: Never grant global edit capabilities if an agent only needs to read or research.

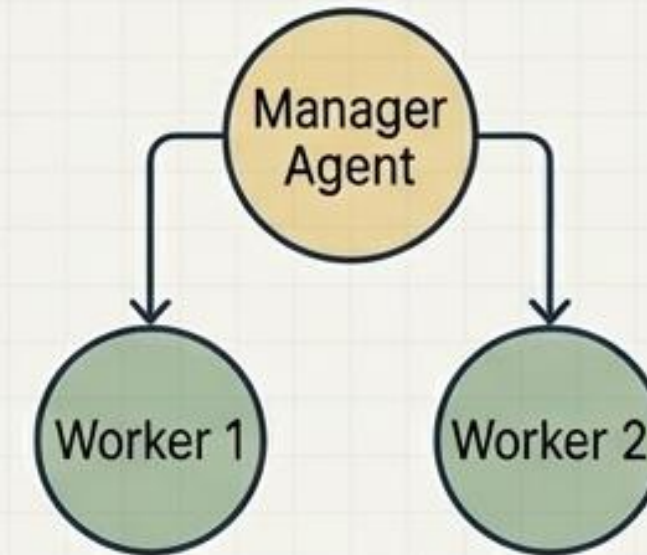
ORCHESTRATION MATRIX: HANDOFFS VS. SUB-AGENTS

HANDOFFS (THE RELAY RACE)



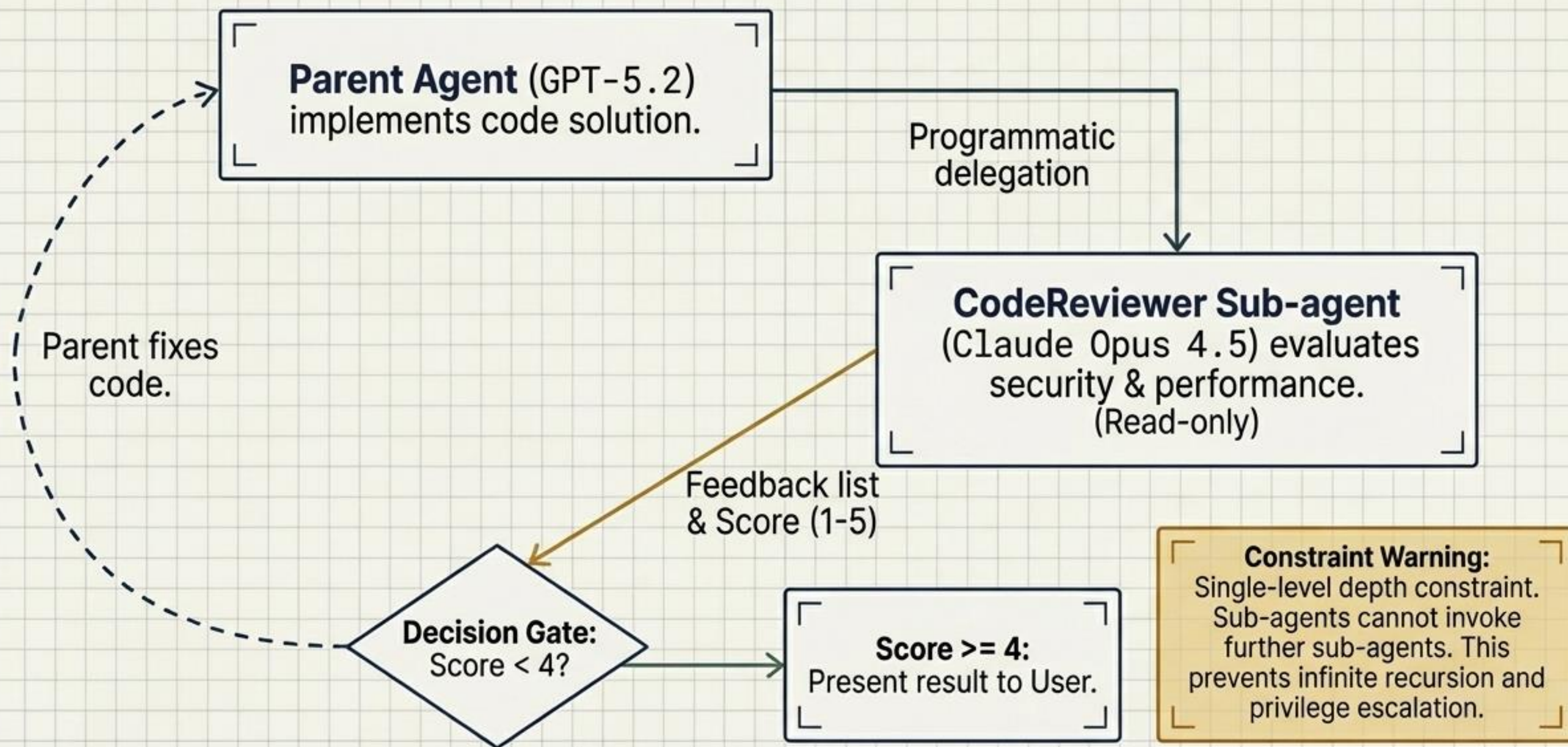
- **TRIGGER:** User clicks a UI button (send: false)
- **CONTROL:** Human-in-the-loop (User-driven)
- **CONTEXT:** Shared conversation history carries over
- **BEST FOR:** Multi-stage pipelines requiring human review (Plan -> Implement -> Review)

SUB-AGENTS (TASK DELEGATION)



- **TRIGGER:** Parent agent invokes programmatically via tools: ['agent']
- **CONTROL:** Agent-autonomous. Transparent to the user.
- **CONTEXT:** Isolated context (clean new session)
- **BEST FOR:** Automated divide-and-conquer tasks and parallel independent runs.

Advanced Pattern: The Automated Review Loop



HOOKS FOR ZERO-TOKEN EXECUTION



The Zero-Token Advantage

Hooks are deterministic scripts configured in `.vscode/settings.json`. They execute locally and bypass the LLM cloud entirely. Therefore, **they consume exactly 0 tokens.**

Note: Copilot uses event-level hooks, differentiating from Claude Code's tool-level hooks.

The UI The Ultimate Anti-Pattern



Token Waste

Global instructions are always-on. Orchestration logic pollutes the context window for every simple, unrelated question.



Context Confusion

Violates separation of concerns. Instructions define personality and global standards; they are not procedural runbooks.



Execution Unreliability

LLMs are non-deterministic. Relying on prompts for complex procedural dispatch frequently causes the agent to hallucinate or deviate.

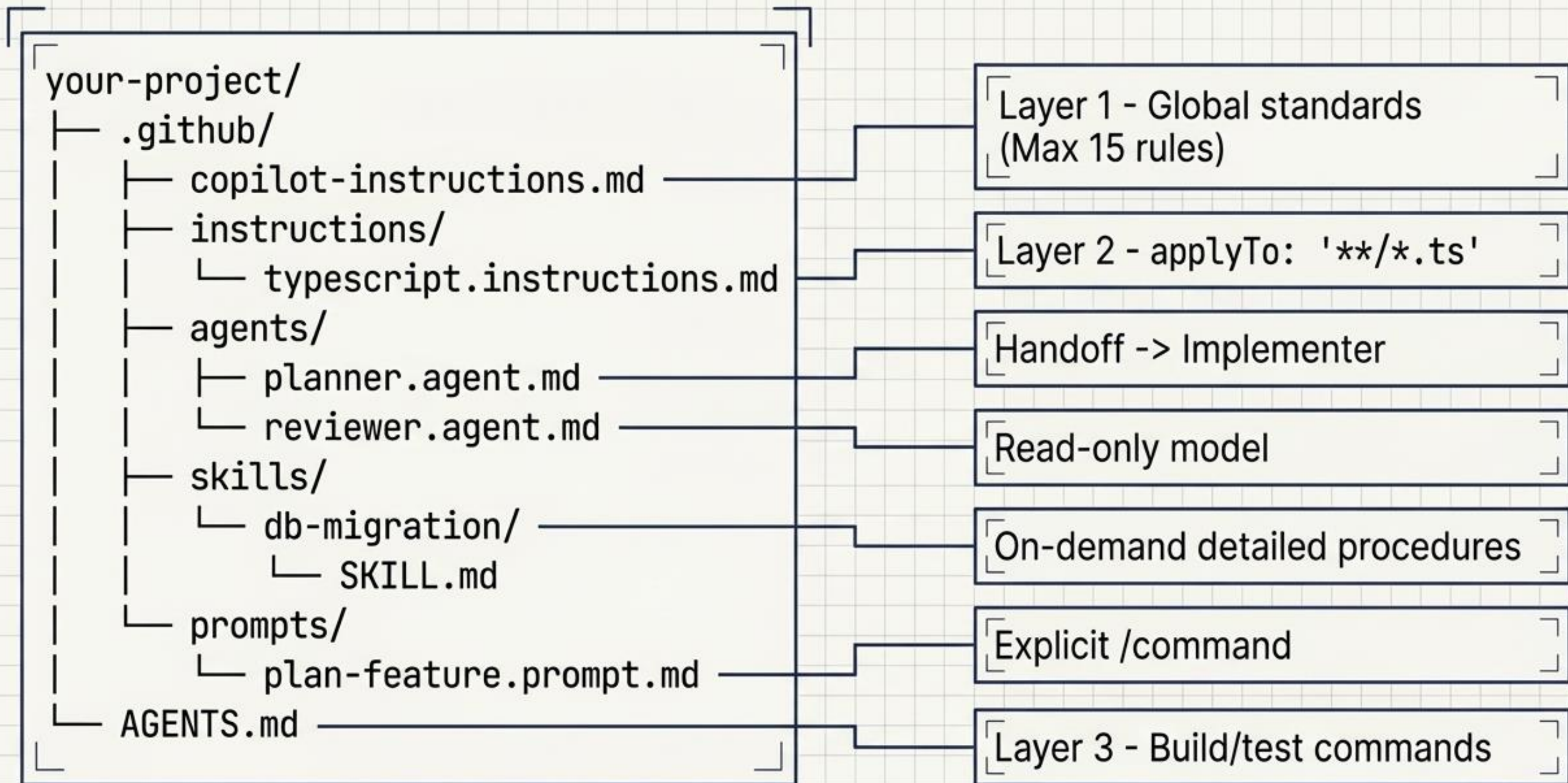


Where Does the Workflow Belong?

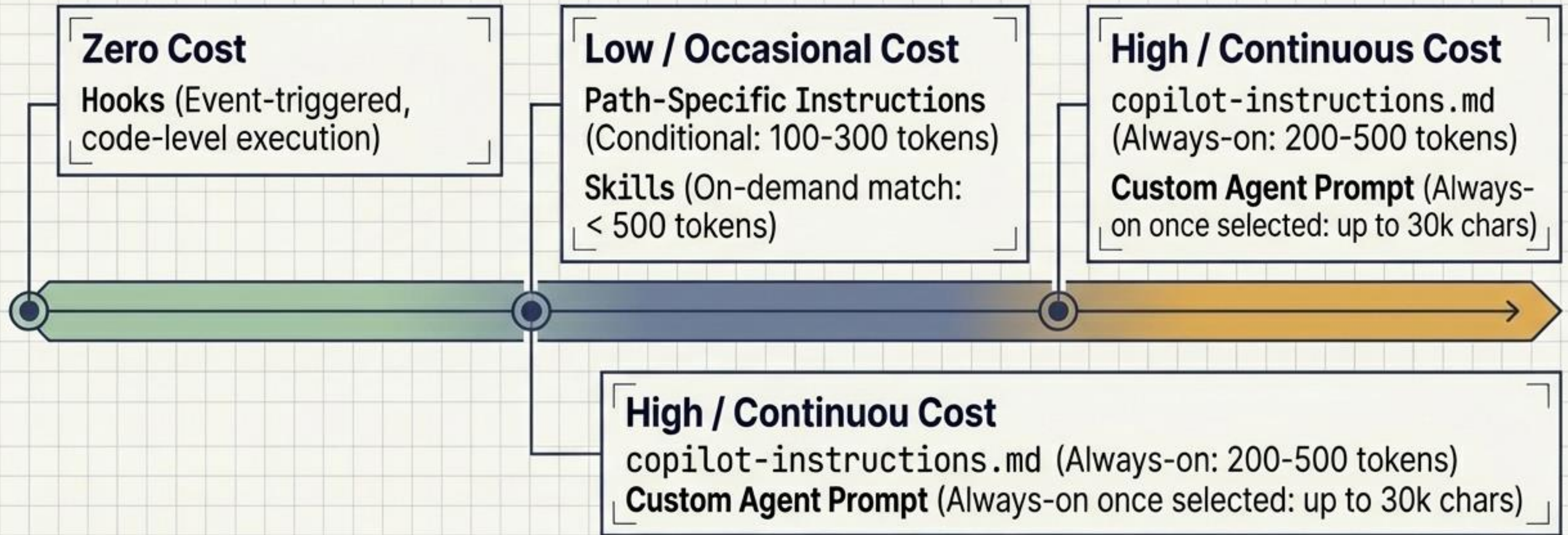
Workflow Type	Correct File Location
Global coding standards (e.g., TS strict mode)	copilot-instructions.md (Always-on)
Simple linear flows (Build -> Test -> Lint)	AGENTS.md (Operational guidance)
Multi-stage pipelines requiring human review	.agent.md + Handoffs (Deterministic UI gating)
Automated divide-and-conquer / Review loops	.agent.md + Sub-agents (Programmatic delegation)
Task-specific detailed procedures (CI debugging)	SKILL.md (On-demand payload)
User-triggered repeatable tasks	.prompt.md (Explicit slash commands)

Synthesis Rule: Never put in Instructions what belongs in a **Skill**.
Never use Instructions when **zero-token Hooks suffice**.

The Perfect Project Directory



Token Cost Strategy Spectrum



Strategic Principle: Never always-on what can be loaded on-demand. Keep global instructions stable to preserve caching.

Security by Design

Default Deny

⚠️ SDK agents cannot execute shell, read files, or fetch URLs by default. Operations require explicit handler approval.

Least Privilege

✓ The custom agent tools whitelist must be restricted. (e.g., Code Reviewers do not need edit or shell access).

Container Isolation

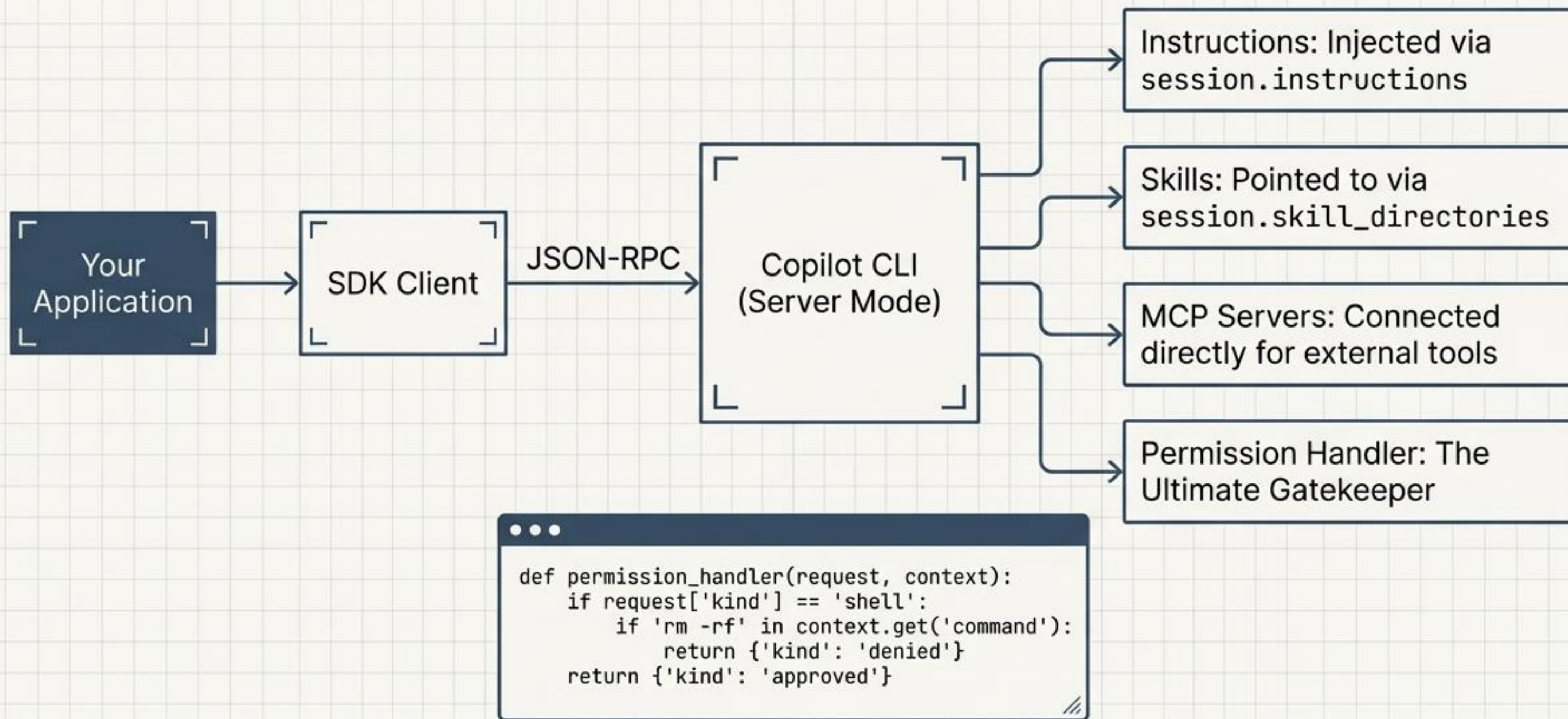
 Agents requiring shell permissions must run inside isolated Dev Containers or Docker instances.

Audit Trails

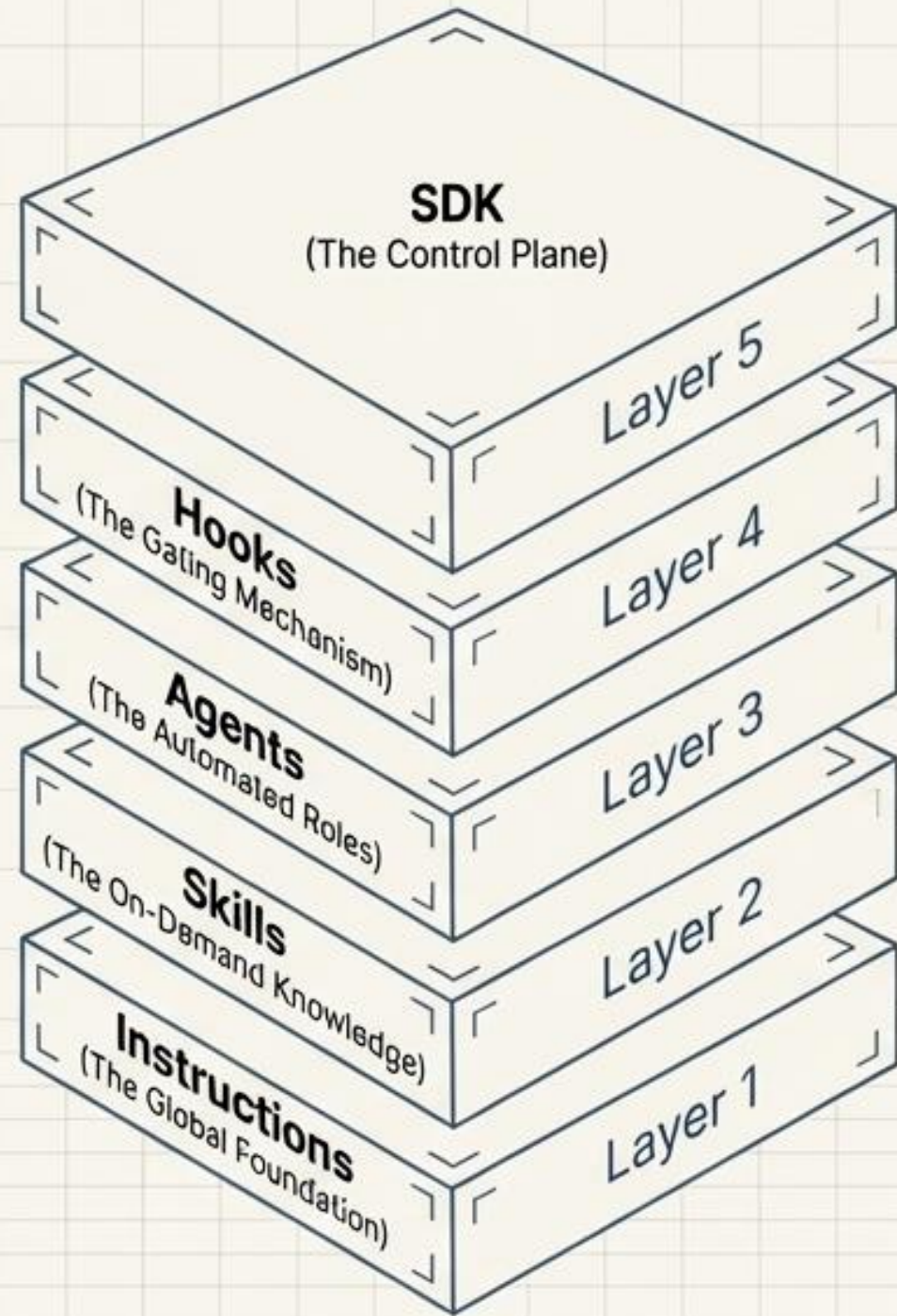
 Utilize PostToolUse hooks to deterministically log every tool invocation for post-hoc security auditing.



The SDK Control Board



The Complete Architectural Blueprint



SDK: Provides remote control. Assemble components programmatically and lock down execution via **Permission Handlers**.

Hooks: Enforce safety deterministically. Offload linting and permission gating here for **zero token cost**.

Agents: Provide automated roles. Use **Handoffs** for human-in-the-loop review, and **Sub-agents** for automated delegation.

Skills: Provide on-demand knowledge. Master **progressive disclosure**; description precision is your highest leverage.

Instructions: Provide foundation. Keep them global and stable. **Never orchestrate workflows here.**